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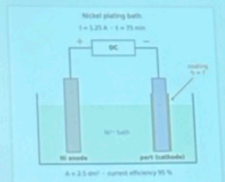
EXAMPLE 1 · QUESTION

Thickness of a nickel coating

Next: Faraday's law plus efficiency and density, to turn a current into a coating thickness.

A steel part with a total surface area of 2.5 dm² is nickel-plated. A constant current of 1.25 A is passed for 75 minutes. The current efficiency of the bath is 95 %. Assume the nickel deposits uniformly over the whole surface.



Given
- area A = 2.5 dm²
- current I = 1.25 A, time t = 75 min
- current efficiency η = 95 %
- Ni: M = 58.7 g/mol, z = 2, density ρ = 8.9 g/cm³
- F = 96 485 C/mol

Find
(a) the mass of nickel deposited (g)
(b) the coating thickness (µm)

NOTE: In plating we want metal to grow on the part, so the part must receive electrons. The negative terminal pushes electrons out, so the part goes to the negative terminal and becomes the cathode. The nickel plate on the positive terminal loses electrons and dissolves: it is the anode. The names come from the reaction, not from the sign.

EXAMPLE 1 · SOLUTION

Thickness of a nickel coating

Step by step: effective current → mass (Faraday) → volume (density) → thickness (area).

Data: area A = 2.5 dm², current I = 1.25 A, time t = 75 min, efficiency η = 0.95, Ni molar mass M = 58.7 g/mol, z = 2, density ρ = 8.9 g/cm³, F = 96 485 C/mol.

Step 1 · Effective current $I_{eff} = \eta I$: only 95 % of the current deposits Ni

$$I_{eff} = \eta I = 0.95 \times 1.25 = 1.1875 \text{ A}$$

Step 2 · Time in seconds

$$t = 75 \times 60 = 4500 \text{ s}$$

Step 3 · Mass of nickel from Faraday's law with I_{eff}

$$m = \frac{M I_{eff} t}{zF} = \frac{58.7 \times 1.1875 \times 4500}{2 \times 96485} = \mathbf{1.626\ g}$$

Step 4 · Volume V = mass / density (cm³)

$$V = \frac{m}{\rho} = \frac{1.626}{8.9} = 0.1827 \text{ cm}^3$$

Step 5 · Area in cm² (1 dm² = 100 cm²)

$$A = 2.5 \text{ dm}^2 = 250 \text{ cm}^2$$

Step 6 · Thickness h = volume / area (1 cm = 10⁴ µm)

$$h = \frac{V}{A} = \frac{0.1827}{250} = 7.31 \times 10^{-4} \text{ cm} = \mathbf{7.3\ \mu m}$$

(a) m = 1.63 g Ni **(b)** h = 7.3 µm

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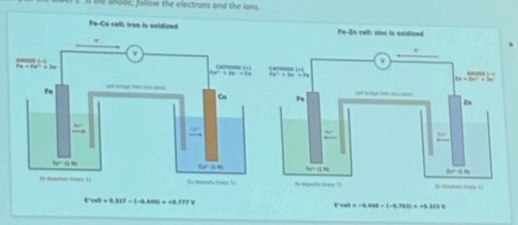
EXAMPLE 6

Two cells with iron: Fe-Cu and Fe-Zn

Step 1: in each pair the lower E° is the anode; follow the electrons and the ions.

- Fe in 1 M Fe^{2+} connected to Cu in 1 M Cu^{2+}
- Fe in 1 M Fe^{2+} connected to Zn in 1 M Zn^{2+}



- **Given:** 25 °C; E°: Cu^{2+}/Cu +0.337 V, Fe^{2+}/Fe −0.440 V, Zn^{2+}/Zn −0.763 V; F = 96485 C/mol
- **Find:** (a) anode and cathode; (b) half-reactions; (c) E°_{cell}; (d) ΔG°; (e) is iron corroded or protected?

- **Fe-Cu:** $E^\circ(Fe) = -0.440 < E^\circ(Cu) = +0.337$, so iron is the **anode** and dissolves; copper deposits.
- **Fe-Zn:** $E^\circ(Zn) = -0.763 < E^\circ(Fe) = -0.440$, so zinc is the anode; iron is the **cathode** and is protected.

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

EXAMPLE 6 · SOLUTION

Two cells with iron: E° and ΔG°

Step 2: the two formulas from "Using the emf series" give E°cell and ΔG° for each cell.

Data: $E^\circ(Cu^{2+}/Cu) = +0.337$ V, $E^\circ(Fe^{2+}/Fe) = -0.440$ V, $E^\circ(Zn^{2+}/Zn) = -0.763$ V; z = 2 electrons; F = 96 485 C/mol.

Cell 1 · (b) anode (oxidation) | cathode (reduction)

$$Fe \rightarrow Fe^{2+} + 2e^- \qquad Cu^{2+} + 2e^- \rightarrow Cu$$

Cell 1 · (c) $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$$E^\circ_{cell} = 0.337 - (-0.440) = \mathbf{+0.777\ V}$$

Cell 1 · (d) $\Delta G^\circ = -zFE^\circ_{cell}$

$$\Delta G^\circ = -2 \times 96\,485 \times 0.777 = \mathbf{-150\ kJ/mol}$$

Cell 2 · (b) anode (oxidation) | cathode (reduction)

$$Zn \rightarrow Zn^{2+} + 2e^- \qquad Fe^{2+} + 2e^- \rightarrow Fe$$

Cell 2 · (c) $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$

$$E^\circ_{cell} = -0.440 - (-0.763) = \mathbf{+0.323\ V}$$

Cell 2 · (d) $\Delta G^\circ = -zFE^\circ_{cell}$

$$\Delta G^\circ = -2 \times 96\,485 \times 0.323 = \mathbf{-62.3\ kJ/mol}$$

(e) Fe-Cu: iron **corrodes** (a scratch in a Cu coating pits the steel). Fe-Zn: iron is **protected** (principle of galvanizing).

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

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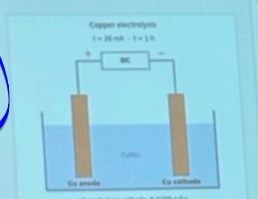
EXAMPLE 2 · QUESTION

Current efficiency of a copper cell

Now the reverse use of efficiency: from a measured mass, find how efficient the cell was.

A current of 26 mA is passed through a copper sulfate solution for exactly 1 hour. When the cathode is weighed afterwards, 0.0300 g of copper has been deposited on it.



Given
- current I = 26 mA, time t = 1 h
- measured mass m = 0.0300 g Cu
- Cu: M = 63.5 g/mol, z = 2
- F = 96 485 C/mol

Find
(a) the current that would deposit 0.0300 g at 100 % efficiency
(b) the current efficiency η of the cell
(c) the mass expected at 100 % efficiency

EXAMPLE 2 · SOLUTION

Current efficiency of a copper cell

Solve Faraday's law for the current, then compare it with the current that was actually used.

Data: actual current I_{actual} = 26 mA = 0.026 A, time t = 1 h, measured mass m = 0.0300 g Cu, molar mass M = 63.5 g/mol, z = 2, F = 96 485 C/mol.

Step 1 · Time in seconds

$$t = 1 \text{ h} = 3600 \text{ s}$$

(a) I_{theo}: the current that gives 0.0300 g at 100 %

$$I_{theo} = \frac{mzF}{Mt} = \frac{0.0300 \times 2 \times 96\,485}{63.5 \times 3600} = \mathbf{25.32\ mA}$$

(b) $\eta = I_{theo} / I_{actual}$

$$\eta = \frac{I_{theo}}{I_{actual}} = \frac{25.32}{26} = \mathbf{0.974}$$

(c) $m_{100\%}$: Faraday mass for the actual current

$$m_{100\%} = \frac{MIt}{zF} = \frac{63.5 \times 0.026 \times 3600}{2 \times 96\,485} = \mathbf{0.0308\ g}$$

Check by mass: $\eta = m_{measured} / m_{Faraday}$

$$\eta = \frac{0.0300}{0.0308} = 0.974$$

The missing 2.6 % of the charge went into side reactions (mainly H_2 evolution) and losses.

(a) 25.3 mA **(b)** η = 0.974 (97.4 %) **(c)** 0.0308 g

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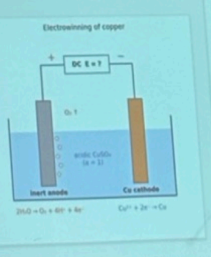
EXAMPLE 7 · QUESTION

Electrowinning of copper

An electrolytic cell that needs both halves of today: the emf series (voltage) and Faraday's law (mass).

Copper is recovered from an acidic $CuSO_4$ solution by electrolysis between an inert anode and a copper cathode (25 °C, all activities = 1). Copper deposits at the cathode and oxygen is evolved at the anode. The cell runs at 200 A.



Given
- Cu^{2+}/Cu: $E^\circ = +0.337$ V
- O_2/H_2O: $E^\circ = +1.229$ V ($O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$)
- current I = 200 A; Cu: M = 63.55 g/mol, z = 2
- F = 96 485 C/mol

Find
(a) the overall cell reaction
(b) the spontaneous (galvanic) direction and its E°_{cell}
(c) the minimum voltage that must be applied
(d) the mass of copper won in 1 h at 100 % efficiency

Note: E° is never multiplied by the coefficients. Multiplying a half-reaction doubles ΔG and z together, so $E = -\Delta G/(zF)$ stays the same. Always use E°cell = E°cathode - E°anode with the table values unchanged.

Take z from the half-reaction of the substance you are asked about. For copper, Cu^{2+} plus two electrons gives Cu, so z equals 2. The four electrons in the overall reaction make two copper atoms, not one.

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

$Cu^{+2} + 2e^- \rightarrow Cu$ (x2) → 4e⁻ — e⁻ sayısını eşitlemek için "Cu" reaksiyonunu 2 ile çarparız.

$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

$\rightarrow 2Cu^{+2} + 2H_2O \rightarrow 2Cu + O_2 + 4H^+$ (a)

Eğer dışarıdan elektrik vermeseydik ve sistem kendi kendine (galvanik) çalışsaydı potansiyeli büyük olan oksijen reaksiyonu (+1.229V) katot, düşük olan bakır (+0.337V) anot olurdu. (DC)

(b) $E^\circ_{cell} = E^\circ_{katot} - E^\circ_{anot} \Rightarrow E_{cell} = 1.229 - 0.337 = +0.892$ V (Spontaneous)

(c) Biz bakırı çözmek için değil kazanmak istediğimiz için sistemi tersine çalışmaya zorluyoruz (elektroliz). Bakır katot, oksijen anot yaptığımızda H potansiyeli negatif kalır. (0.337 − 1.229 = −0.892V). Bu negatif potansiyeli yok reaksiyonu başlatabilmek için dışarıdaki DC güç kaynağının sisteme en az +0.892V gerilim uygulanmalıdır.

+0.892V min Voltage.

(D) Faraday kanununu kullanarak kütleyi bulurız. (%100 efficiency)

I = 200A, t = 3600 s, M = 63.55 g/mol, F = 96485 C/mol

Toplam reaksiyonda 4e⁻ görünsede Cu iyonu (Cu^{+2}) metalik bakıra dönüşürken 2e⁻ alır → z = 2

$m = \frac{MIt}{zF} = \frac{63.55 \times 200 \times 3600}{2 \times 96485} = 237.11$ g

Sistem %100 verimle çalıştığında 1 saatte katottan 237.11 gram bakır biriktir.

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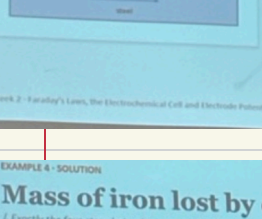
EXAMPLE 4 · QUESTION

Mass of iron lost by corrosion

The same law in corrosion: here metal is lost at the anode instead of being plated at a cathode.

On a corroding steel surface the anodic reaction is $Fe \rightarrow Fe^{2+} + 2e^-$. A constant current of 0.50 mA flows between the anodic and cathodic areas for one year (365 days).



Given
- current I = 0.50 mA = 5.0 × 10^{-4} A
- time t = 365 days
- Fe: M = 55.85 g/mol, z = 2
- F = 96 485 C/mol, $N_A = 6.022 \times 10^{23}$ mol^{-1}

Find
(a) the charge passed in one year (C)
(b) the moles of electrons released
(c) the mass of iron lost (g)
(d) the number of iron atoms lost

NOTE: The current between the anodic and cathodic areas is called the corrosion current. If we know it, Faraday's law tells us how much iron is lost.

Week 2 · Faraday's Laws, the Electrochemical Cell and Electrode Potential

EXAMPLE 4 · SOLUTION

Mass of iron lost by corrosion

Exactly the four-step chain: charge → moles of electrons → moles of Fe → grams (and atoms).

Data: current I = 5.0 × 10^{-4} A, time t = 365 days, Fe: molar mass M = 55.85 g/mol, z = 2, F = 96 485 C/mol, Avogadro's number N_A = 6.022 × 10^{23} mol^{-1}.

Step 1 · Time in seconds

$$t = 365 \times 24 \times 3600 = 3.154 \times 10^7 \text{ s}$$

(a) Charge Q = I·t

$$Q = It = 5.0 \times 10^{-4} \times 3.154 \times 10^7 = 1.577 \times 10^4 \text{ C}$$

(b) Moles of electrons $n_e = Q/F$

$$n_e = \frac{Q}{F} = \frac{15\,768}{96\,485} = 0.1634 \text{ mol}$$

Step 4 · Moles of iron $n_{Fe} = n_e/2$

$$n_{Fe} = \frac{n_e}{2} = 0.0817 \text{ mol}$$

(c) Mass $m = n_{Fe} \times M$

$$m = n_{Fe} M = 0.0817 \times 55.85 = \mathbf{4.56\ g}$$

(d) Atoms $N = n_{Fe} \times N_A$

$$N = n_{Fe} N_A = 0.0817 \times 6.022 \times 10^{23} = \mathbf{4.92 \times 10^{22}}$$

5×10^{-4} · 365 × 24 × 3600 → $m = \frac{MIT}{zF}$ → 96485 · 2 · 55.85

(a) 1.58×10^4 C **(b)** 0.163 mol e^- **(c)** 4.56 g Fe **(d)** 4.92×10^{22} atoms

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EXAMPLE 1 · QUESTION

Standard potentials from free-energy data

Practise both ideas together: E° from chemical potentials, and K from ΔG°.

Use the standard chemical potentials below to calculate, at 25 °C, the standard reduction potentials of the Zn^{2+}/Zn and Cu^{2+}/Cu electrodes, and then the standard potential and the equilibrium constant of the Daniell cell $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$.

Given
- $\mu^\circ(Zn^{2+}) = -147.1$ kJ/mol
- $\mu^\circ(Cu^{2+}) = +65.5$ kJ/mol
- $\mu^\circ(Zn) = \mu^\circ(Cu) = 0$
- F = 96485 C/mol, R = 8.314 J/(mol·K), T = 298.15 K

Anot Katot

Note: (ΔG° in joules)

Find
(a) E° of $Zn^{2+} + 2e^- \rightarrow Zn$
(b) E° of $Cu^{2+} + 2e^- \rightarrow Cu$
(c) ΔG° and E° of the Daniell cell
(d) the equilibrium constant K of the cell reaction

Week 3 · Thermodynamics of Corrosion and the Causes of Potential Differences

EXAMPLE 1 · SOLUTION (1/2)

The two half-cells

Step by step: write each half-reaction as a reduction, find ΔG°, divide by −zF.

Data: $\mu^\circ(Zn^{2+}) = -147.1$ kJ/mol, $\mu^\circ(Cu^{2+}) = +65.5$ kJ/mol, $\mu^\circ(Zn) = \mu^\circ(Cu) = 0$; z = 2; F = 96 485 C/mol.

ΔG° = Ürünler − Girenler

(a) Step 1 · ΔG° of the zinc reduction = $\mu^\circ(Zn) - \mu^\circ(Zn^{2+})$

$$Zn^{2+} + 2e^- \rightarrow Zn: \quad \Delta G^\circ = 0 - (-147.1) = +147.1 \text{ kJ}$$

(a) Step 2 · E° = −ΔG° / zF (ΔG° in joules)

$$E^\circ_{Zn} = -\frac{147\,100}{2 \times 96\,485} = \mathbf{-0.762\ V} \qquad E^\circ = \frac{-\Delta G^\circ \text{ (joule)}}{zF}$$

(b) Step 1 · ΔG° of the copper reduction = $\mu^\circ(Cu) - \mu^\circ(Cu^{2+})$

$$Cu^{2+} + 2e^- \rightarrow Cu: \quad \Delta G^\circ = 0 - 65.5 = -65.5 \text{ kJ}$$

(b) Step 2 · E° = −ΔG° / zF

$$E^\circ_{Cu} = -\frac{-65\,500}{2 \times 96\,485} = \mathbf{+0.339\ V}$$

(a) $E^\circ(Zn^{2+}/Zn) = -0.762$ V **(b)** $E^\circ(Cu^{2+}/Cu) = +0.339$ V (table: −0.763 V and +0.337 V)

EXAMPLE 1 · SOLUTION (2/2)

The Daniell cell and its equilibrium constant

Now the whole cell: add the two reactions; ΔG° then gives both E°cell and the equilibrium constant.

Cell reaction: $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$ (z = 2). Data: as before; R = 8.314 J/(mol·K), T = 298.15 K.

(c) ΔG° of the cell = $\mu^\circ(Zn^{2+}) + \mu^\circ(Cu) - \mu^\circ(Zn) - \mu^\circ(Cu^{2+})$, then E°cell = −ΔG° / zF

$$\Delta G^\circ = -147.1 - 65.5 = -212.6 \text{ kJ} \qquad E^\circ_{cell} = \frac{212\,600}{2 \times 96\,485} = \mathbf{+1.10\ V}$$

(d) Equilibrium constant from $\ln K = -\Delta G^\circ / RT$

$$\ln K = \frac{-\Delta G^\circ}{RT} = \frac{212\,600}{8.314 \times 298.15} = 85.8 \qquad K = \mathbf{1.8 \times 10^{37}}$$

Check: E°cell = E°cathode − E°anode = 0.339 − (−0.762) = 1.10 V, the same result.

Meaning of K: at equilibrium $a(Zn^{2+}) / a(Cu^{2+}) \approx 10^{37}$. The reaction runs until practically no Cu^{2+} is left.

Such a huge K means the reaction goes practically to completion.

ΔG° = −212.6 kJ, E°cell = +1.10 V **(d)** $K \approx 1.8 \times 10^{37}$

EXAMPLE · QUESTION

Electrode potentials away from 1 M

First practice with Nernst: one electrode at a time, forwards and backwards.

Three metal electrodes are placed in solutions of their own ions at 25 °C. Calculate the electrode potentials in (a) and (b) and the ion activity in (c). Take activities equal to concentrations.

Given

- $E°(Zn^{2+}/Zn) = -0.763$ V
- $E°(Cu^{2+}/Cu) = +0.337$ V
- $E°(Fe^{2+}/Fe) = -0.440$ V
- $z = 2$ (Zn^{2+}, Cu^{2+}, Fe^{2+}); 25 °C

Find

(a) E of zinc in 0.010 M Zn^{2+}

(b) E of copper in 1.0×10^{-4} M Cu^{2+}

(c) the Fe^{2+} activity at which an iron electrode reads E = −0.500 V (vs SHE)

$E = E° - \frac{RT \ln Q}{zF}$

$E = E° - \frac{0.0592}{z} \log Q$ at 25 °C

$Zn^{+2} + 2e^- \rightarrow Zn$

$Q = \frac{a_{Zn}}{a_{Zn^{+2}}}$

$E = E° + 0.0592 \log a_{Zn^{+2}}$

EXAMPLE 2 · SOLUTION

Electrode potentials away from 1 M

Each answer is E° plus 0.0296 times the log of the activity; in (c) we solve for the log first.

(a) E of zinc in 0.010 M Zn^{2+}
(b) E of copper in 1.0×10^{-4} M Cu^{2+}
(c) the Fe^{2+} activity at which an iron electrode reads E = −0.500 V (vs SHE)

Data: $E°(Zn) = -0.763$ V, $E°(Cu) = +0.337$ V, $E°(Fe) = -0.440$ V; $z = 2$, so $0.0592/z = 0.0296$ V; 25 °C.

$E = E° - \frac{RT}{zF} \ln Q$ at 25 °C: $E = E° - \frac{0.0592}{z} \log Q$

(a) Zinc in 0.010 M: $\log(0.010) = -2$ — $E_{Zn} = -0.763 + \frac{0.0592}{2} \log(0.010) = -0.763 - 0.059 = \mathbf{-0.822\ V}$

(b) Copper in 1.0×10^{-4} M: $\log(10^{-4}) = -4$ — $E_{Cu} = 0.337 + \frac{0.0592}{2} \log(1.0 \times 10^{-4}) = 0.337 - 0.118 = \mathbf{+0.219\ V}$

(c) Step 1 · put E = −0.500 V into the equation and solve for log a — $-0.500 = -0.440 + 0.0296 \log a \Rightarrow \log a = -2.03$

(c) Step 2 · take the antilog — $a_{Fe^{2+}} = 10^{-2.03} = \mathbf{9.4 \times 10^{-3}}$ (≈ 0.0094 M)

(a) −0.822 V **(b)** +0.219 V **(c)** $a(Fe^{2+}) \approx 9.4 \times 10^{-3}$

EXAMPLE · QUESTION

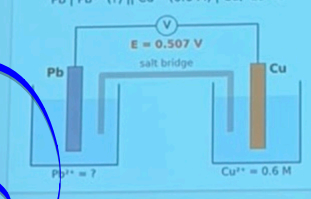
Unknown ion concentration from a cell voltage

Next, a whole cell with an unknown concentration: the classic exam question.

An electrochemical cell consists of pure copper and pure lead electrodes immersed in solutions of their respective divalent ions. The Cu^{2+} concentration is 0.6 M. The lead electrode is oxidized and the cell potential is 0.507 V at 25 °C. Calculate the Pb^{2+} concentration.

Pb yükseltgenecek. Bakır oluşacak.

$Pb \mid Pb^{2+}\ (?) \parallel Cu^{2+}\ (0.6\ M) \mid Cu,\ 25\ °C$



Given

- $[Cu^{2+}] = 0.6$ M; Pb is oxidized
- $E_{cell} = 0.507$ V, T = 25 °C
- $E°(Cu^{2+}/Cu) = +0.337$ V
- $E°(Pb^{2+}/Pb) = -0.126$ V
- activities = concentrations

Find

(a) the cell reaction and z

(b) E°cell

(c) $[Pb^{2+}]$ (mol/L)

EXAMPLE 3 · SOLUTION

Unknown ion concentration from a cell voltage

Reaction first, then E°cell, then Nernst solved for the unknown.

(a) the cell reaction and z
(b) E°cell
(c) $[Pb^{2+}]$ (mol/L)

Data: $[Cu^{2+}] = 0.6$ M, E = 0.507 V, $E°(Cu) = +0.337$ V, $E°(Pb) = -0.126$ V, 25 °C; lead is the anode.

(a) and (b) · Pb is oxidized, Cu^{2+} is reduced; E°cell = E°cathode − E°anode — $Pb + Cu^{2+} \rightarrow Pb^{2+} + Cu \quad (z = 2) \qquad E°_{cell} = 0.337 - (-0.126) = 0.463$ V

Step 3 · Nernst for the cell: Q = $[Pb^{2+}] / [Cu^{2+}]$ (products over reactants) — $E = E° - \frac{0.0592}{2} \log \frac{[Pb^{2+}]}{[Cu^{2+}]}$ $\quad E = E° - \frac{RT}{zF} \ln Q$ at 25 °C: $E = E° - \frac{0.0592}{z} \log Q$

Step 4 · insert the numbers and solve for the log — $0.507 = 0.463 - 0.0296 \log \frac{[Pb^{2+}]}{0.6} \Rightarrow \log \frac{[Pb^{2+}]}{0.6} = -1.49$

(c) take the antilog and multiply by 0.6 — $[Pb^{2+}] = 0.6 \times 10^{-1.49} = \mathbf{0.020\ M}$

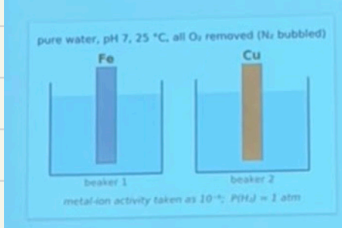
(a) $Pb + Cu^{2+} \rightarrow Pb^{2+} + Cu$, z = 2 **(b)** E°cell = 0.463 V **(c)** $[Pb^{2+}] \approx 0.020$ M

EXAMPLE · QUESTION

Can iron and copper corrode without oxygen?

Back to the opening question: let us calculate whether iron and copper can corrode in oxygen-free water.

Pure iron and pure copper samples are placed in pure water at pH 7 and 25 °C from which all dissolved oxygen has been removed. Take the metal-ion activity at the surface as 10^{-6} and $P(H_2) = 1$ atm. Decide with calculations whether each metal can corrode, then repeat for copper in air-saturated water.



Given

- $E°(Fe^{2+}/Fe) = -0.440$ V
- $E°(Cu^{2+}/Cu) = +0.337$ V
- $E°(O_2/H_2O) = +1.229$ V
- $a(Fe^{2+}) = a(Cu^{2+}) = 10^{-6}$
- pH 7, 25 °C, air: $P(O_2) = 0.21$ atm

Find

(a) E of iron and E of the hydrogen electrode

(b) E_{cell} for $Fe + 2H^+ \rightarrow Fe^{2+} + H_2$; possible?

(c) the same test for copper

(d) copper in air-saturated water (O_2 present)

(a) Demirin ve Hidrojen Elektrodunun Gerçek Potansiyelleri (E)
Burada standart değerleri (E°) değil, Nernst denklemiyle o anki aktivitedeki (10^{-6}) gerçek değerleri buluyoruz. JPG

- **Demir (Fe) için:** $E_{Fe} = E^\circ + \frac{0.0592}{z} \log(a_{Fe^{2+}})$
  Demir 2 elektron verir ($z = 2$).
  $E_{Fe} = -0.440 + \frac{0.0592}{2} \log(10^{-6})$
  $E_{Fe} = -0.440 + 0.0296 \times (-6) = -0.440 - 0.1776 = \mathbf{-0.618\ V}$
- **Hidrojen (H_2) için (Oksijensiz suyun katodu):**
  Önceki slayttaki pratik formülü kullanıyoruz: $E_H = -0.0592 \times \text{pH}$
  $E_H = -0.0592 \times 7 = \mathbf{-0.414\ V}$

(b) Oksijensiz Ortamda Demir Korozyona Uğrar mı? ($Fe + 2H^+ \rightarrow Fe^{2+} + H_2$)
Korozyon olması için hücre potansiyelinin pozitif çıkması ($E_{cell} > 0$) gerekir.
Demir eridiği için anot, sudaki hidrojen gaz çıkardığı için katottur. JPG

- $E_{cell} = E_{katot} - E_{anot}$
- $E_{cell} = E_H - E_{Fe}$
- $E_{cell} = -0.414 - (-0.618) = \mathbf{+0.204\ V}$
  Sonuç: $E_{cell} > 0$ olduğu için reaksiyon spontanedir. **Evet, demir içinde hiç oksijen olmasa bile suda korozyona uğrar (paslanmasa da çözünür).**

(c) Aynı Testi Bakır İçin Yapalım
Önce bakırın o anki gerçek potansiyelini bulalım.

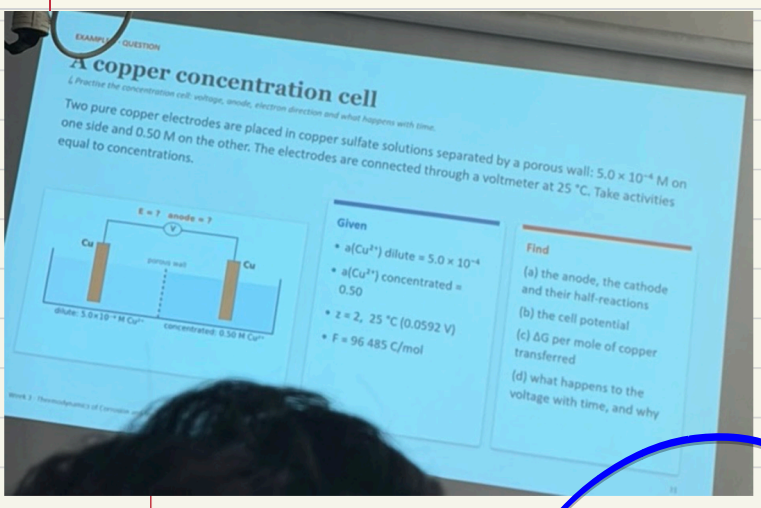
- **Bakır (Cu) için:** $E_{Cu} = E^\circ + \frac{0.0592}{2} \log(a_{Cu^{2+}})$
  $E_{Cu} = +0.337 + 0.0296 \times (-6) = +0.337 - 0.1776 = \mathbf{+0.159\ V}$

Şimdi hidrojene karşı hücre potansiyelini hesaplayalım:

- $E_{cell} = E_H - E_{Cu}$
- $E_{cell} = -0.414 - (+0.159) = \mathbf{-0.573\ V}$
  Sonuç: $E_{cell} < 0$ çıktı. **Hayır, bakır oksijensiz saf suda kesinlikle korozyona uğramaz**

(d) Bakır Havalandırılmış (Oksijenli) Suda Korozyona Uğrar mı?
Ortamda oksijen ($P(O_2) = 0.21$ atm) olduğunda katot artık hidrojen değil, oksijen reaksiyonu olur. JPG

- **Oksijenin Potansiyeli (E_O):** Önceki slayttaki uzun formülden hesaplarsak (pH 7 ve havada) yaklaşık $+0.805$ V çıkar. JPG
- Yeni Hücre Potansiyeli: $E_{cell} = E_{katot}(Oksijen) - E_{anot}(Bakır)$
- $E_{cell} = +0.805 - (+0.159) = \mathbf{+0.646\ V}$
  Sonuç: $E_{cell} > 0$ olduğu için reaksiyon artık spontanedir. **Evet, bakır ancak suya oksijen karıştığında korozyona uğrayabilir.**



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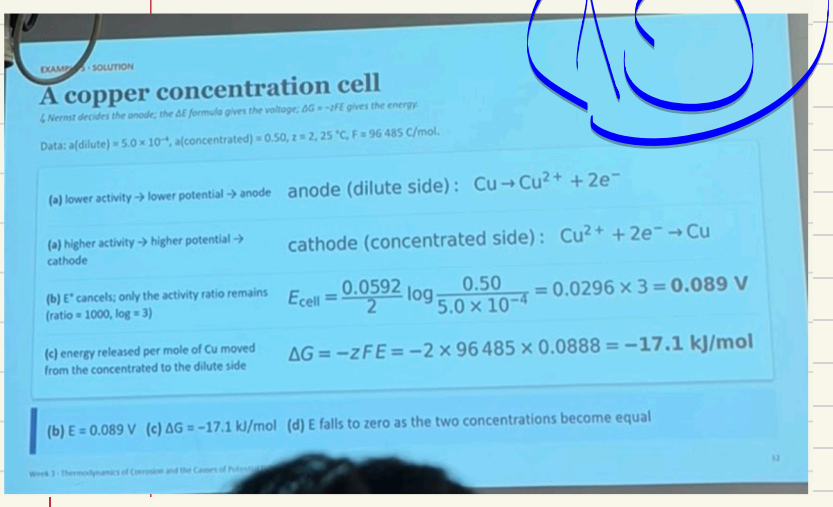
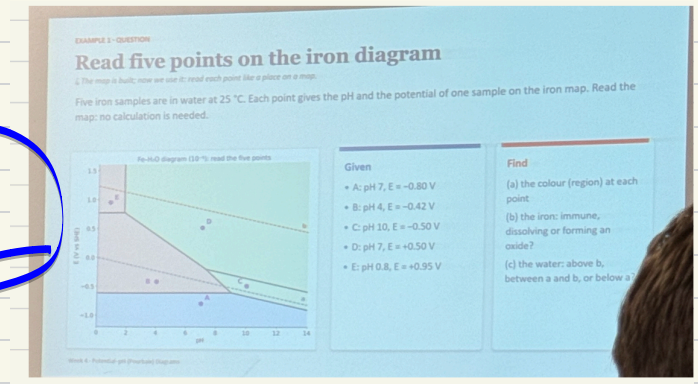




12

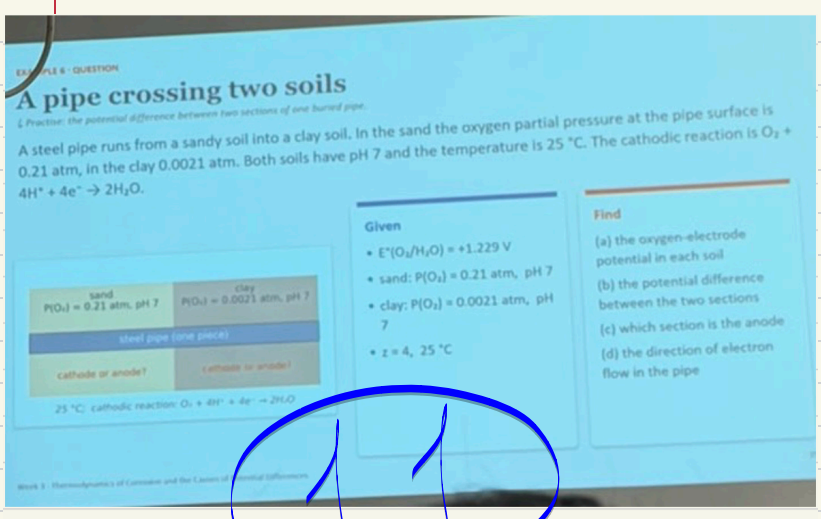
a) Immunity region → Pure IRON,

b) IRON divalent ions , Corossion region

c) Passivity Region , Magnetite

D) Passivity Region, Hematite

E) Corrosion Region, IRON Trivalent Ions



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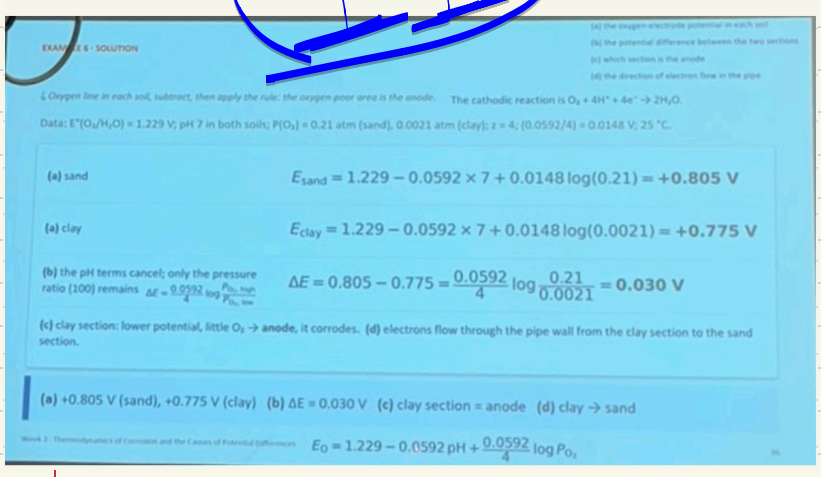
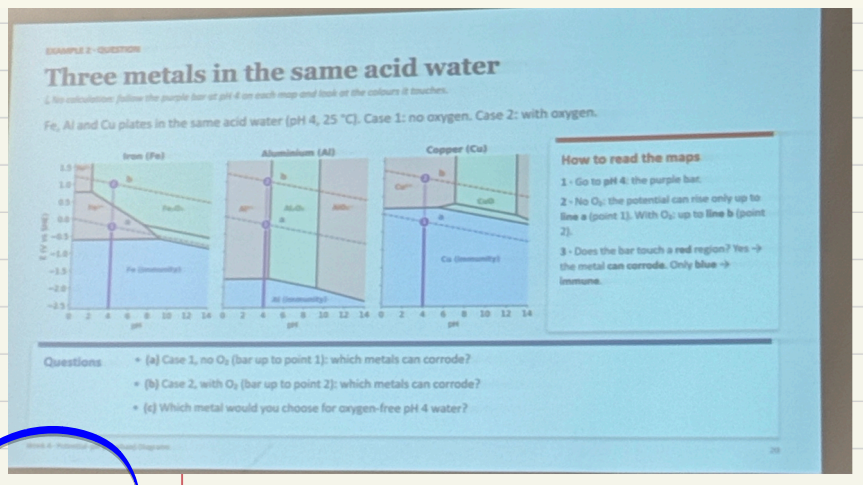


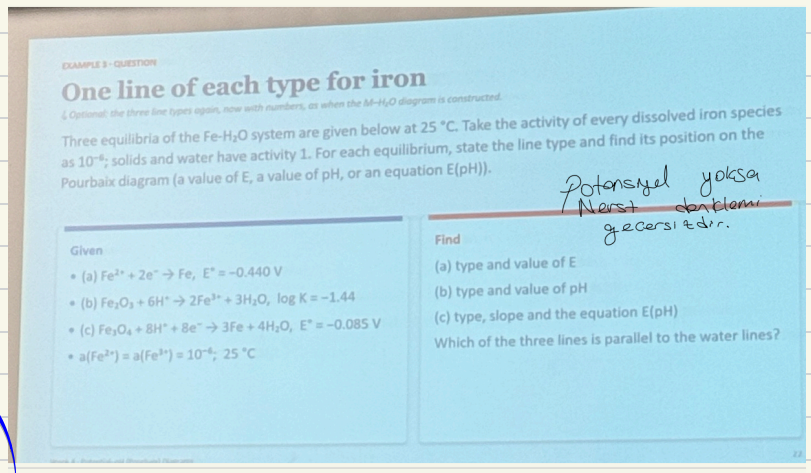


13

a) IRON, Aluminum => "O_2" make Corrosion Copper cant. → Bakır kullanırım

B)

C) I will choose Copper.



Potansiyel yoksa Nerst denklemi geçersizdir.

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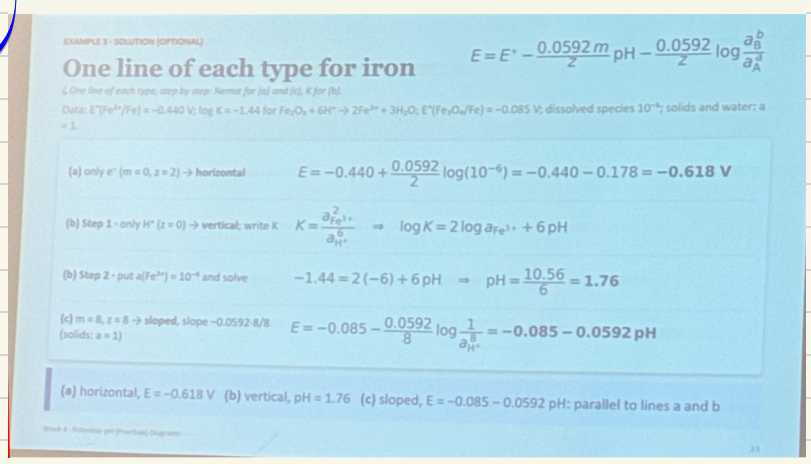
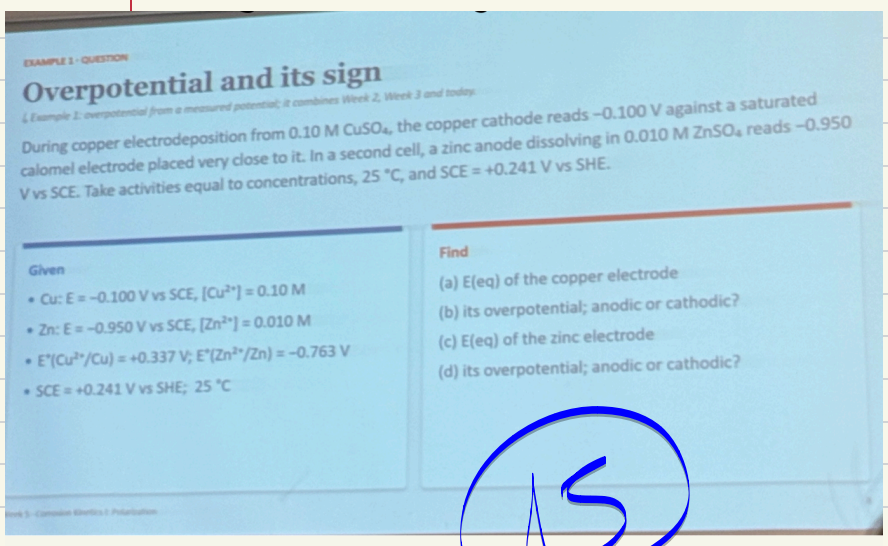




EXAMPLE 1 · QUESTION

Overpotential and its sign

↳ Example 1: overpotential from a measured potential; it combines Week 2, Week 3 and today.

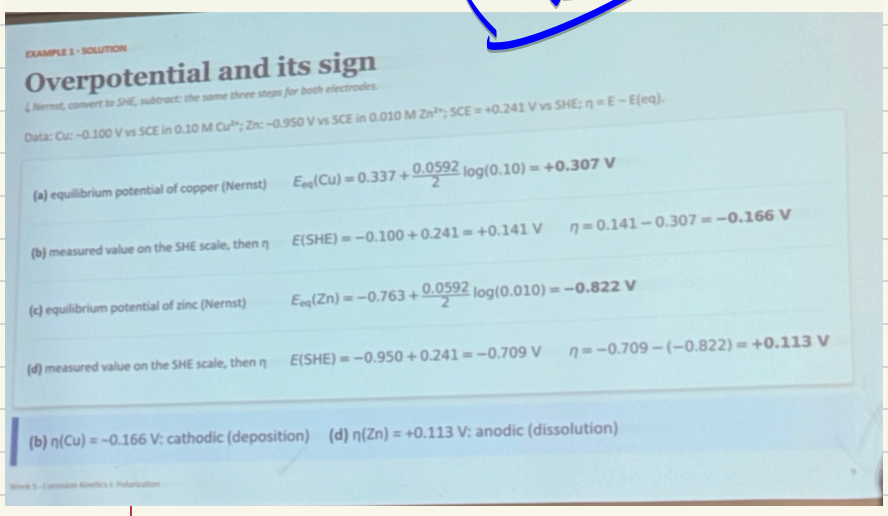
During copper electrodeposition from 0.10 M $CuSO_4$, the copper cathode reads −0.100 V against a saturated calomel electrode placed very close to it. In a second cell, a zinc anode dissolving in 0.010 M $ZnSO_4$ reads −0.950 V vs SCE. Take activities equal to concentrations, 25 °C, and SCE = +0.241 V vs SHE.

Given

- Cu: E = −0.100 V vs SCE, $[Cu^{2+}] = 0.10$ M
- Zn: E = −0.950 V vs SCE, $[Zn^{2+}] = 0.010$ M
- $E°(Cu^{2+}/Cu) = +0.337$ V; $E°(Zn^{2+}/Zn) = -0.763$ V
- SCE = +0.241 V vs SHE; 25 °C

Find

(a) E(eq) of the copper electrode
(b) its overpotential; anodic or cathodic?
(c) E(eq) of the zinc electrode
(d) its overpotential; anodic or cathodic?



EXAMPLE 1 · SOLUTION

Overpotential and its sign

↳ Nernst, convert to SHE, subtract: the same three steps for both electrodes.

Data: Cu: −0.100 V vs SCE in 0.10 M Cu^{2+}; Zn: −0.950 V vs SCE in 0.010 M Zn^{2+}; SCE = +0.241 V vs SHE; $\eta = E - E(eq)$.

(a) equilibrium potential of copper (Nernst) $E_{eq}(Cu) = 0.337 + \frac{0.0592}{2}\log(0.10) = \mathbf{+0.307\ V}$

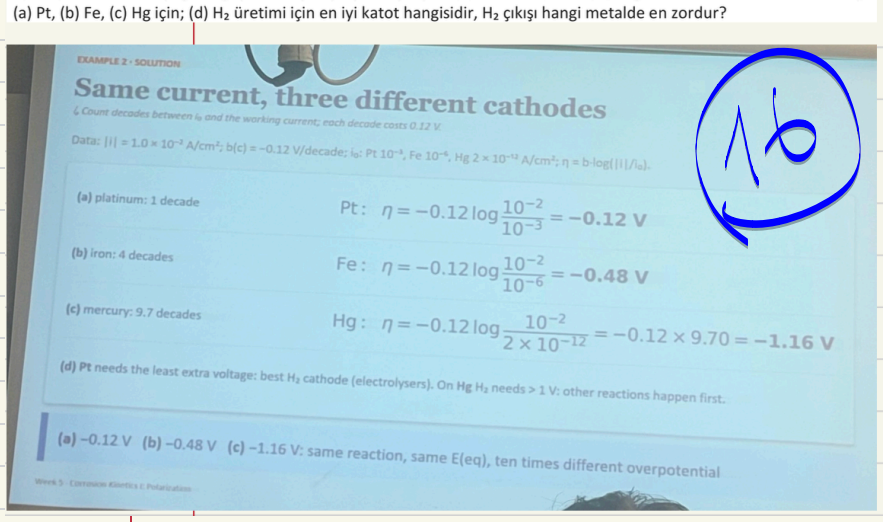
(b) measured value on the SHE scale, then η $E(SHE) = -0.100 + 0.241 = +0.141$ V $\quad \eta = 0.141 - 0.307 = \mathbf{-0.166\ V}$

(c) equilibrium potential of zinc (Nernst) $E_{eq}(Zn) = -0.763 + \frac{0.0592}{2}\log(0.010) = \mathbf{-0.822\ V}$

(d) measured value on the SHE scale, then η $E(SHE) = -0.950 + 0.241 = -0.709$ V $\quad \eta = -0.709 - (-0.822) = \mathbf{+0.113\ V}$

(b) η(Cu) = −0.166 V: cathodic (deposition) **(d)** η(Zn) = +0.113 V: anodic (dissolution)

Soru 3 · Aynı akım, üç farklı katot (derste Örnek 2). Asit içinde platin, demir ve cıva üzerinde 10 mA/cm² (1.0×10^{-2} A/cm²) katodik akım yoğunluğuyla hidrojen çıkıyor. Üçü için b(c) = −0.12 V/dekat, i_0 değerlerini tablodan alın. Bulun: η, (a) Pt, (b) Fe, (c) Hg için; (d) H_2 üretimi için en iyi katot hangisidir, H_2 çıkışı hangi metalde en zordur?



EXAMPLE 2 · SOLUTION

Same current, three different cathodes

↳ Count decades between i_0 and the working current; each decade costs 0.12 V.

Data: $|i| = 1.0 \times 10^{-2}$ A/cm²; b(c) = −0.12 V/decade; i_0: Pt 10^{-3}, Fe 10^{-6}, Hg 2×10^{-12} A/cm²; $\eta = b \cdot \log(|i|/i_0)$.

(a) platinum: 1 decade Pt: $\eta = -0.12 \log \frac{10^{-2}}{10^{-3}} = \mathbf{-0.12\ V}$

(b) iron: 4 decades Fe: $\eta = -0.12 \log \frac{10^{-2}}{10^{-6}} = \mathbf{-0.48\ V}$

(c) mercury: 9.7 decades Hg: $\eta = -0.12 \log \frac{10^{-2}}{2 \times 10^{-12}} = -0.12 \times 9.70 = \mathbf{-1.16\ V}$

(d) Pt needs the least extra voltage: best H_2 cathode (electrolysers). On Hg H_2 needs > 1 V: other reactions happen first.

(a) −0.12 V **(b)** −0.48 V **(c)** −1.16 V: same reaction, same E(eq), ten times different overpotential